

Response to Reviewers

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AI-Powered Unified Mobility and Delivery System Using Dynamic Feasibility Analysis

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We thank both reviewers for the review. The comments identified two problems that went beyond presentation: an experimental record we could not reconstruct, and claims about learning and about data handling that the implementation did not support. We have resolved both by changing the manuscript to match the system rather than by defending the earlier text.

Three points are stated up front so that nothing below is read as more than it is. First, the 43-request pilot and the 2,320-trip deployment figures have been removed in full and replaced by a reproducible 36-run experiment; no number from the withdrawn datasets survives anywhere in the paper. Second, the gradient-based learning rule was never trained and is now labelled throughout as a proposed extension; what is evaluated is the bounded heuristic adaptation the system actually implements. Third, several comments cannot be resolved by any change to the text — per-service metrics, driver and customer acceptance, and a direct comparison against another joint-routing algorithm all require work that has not been done. Those are marked below as acknowledged limitations, not as resolved.

All modifications are highlighted in the revised manuscript.

Reviewer 1

Comment 1: *Only 43 requests from a single bounded area are used by the pilot evaluation, which is too small to provide statistically reliable evidence of the framework's generalizability across different cities, traffic conditions, and service-demand levels.*

Response: We agree, and we have not tried to defend the 43-request pilot. That evaluation has been withdrawn from the manuscript entirely. In its place the camera-ready reports a reproducible experiment over four workload sizes — 50, 100, 250 and 500 requests — with five random seeds at the three smaller sizes and three at 500, giving 36 runs. Every comparison is paired on identical seeded request pools. We want to be exact about what this does and does not settle: it is a scalability evaluation under a controlled synthetic workload, and it does not demonstrate generalisability across cities or traffic regimes. We have not claimed that it does.

Change made: The 43-request pilot and all figures derived from it were removed. Sections IV and V were rewritten around the 36-run experiment. A limitation stating that the workload is synthetic, single-region and without live traffic, and that multi-city validation remains necessary, was added.

Section / location: Section IV-A (p. 5), Section V in full (pp. 5–7), Section VI paragraph 1 (p. 7).

Status: Addressed by replacing the evaluation. The generalisability limitation is retained and stated explicitly, not resolved.

Comment 2: *The proposed adaptive coefficient-learning mechanism is presented as a key AI contribution, but it was not actually trained or validated using real operational outcomes, so its practical effectiveness remains unverified.*

Response: This was correct and the manuscript now separates two different things that the earlier version ran together. What the system implements is a bounded heuristic adaptation: the five scoring weights are regenerated each dispatch cycle from measurable operating context and from per-corridor correction factors refitted on completed-trip residuals, with every learned correction clipped at ± 0.15 .

That mechanism is implemented and it is now evaluated, as a seed-matched paired ablation against the static mode, and separately in a multi-day closed-loop experiment. What the system does not do is train a softmax-parameterised weight vector by gradient descent on a cross-entropy objective. Those relations are retained as mathematics but are labelled throughout as a proposed extension that has not been run. We also report that the learning component was invoked zero times during the 36 scalability runs, because a single dispatch pass produces no completed trip and therefore no residual.

Change made: Section III-D now describes the implemented adaptive mechanism and states plainly that it is not gradient descent. Section III-E is titled “Proposed Learning Extension” and closes by stating that no softmax parameterisation, cross-entropy objective or gradient update was executed in any experiment reported. Section V-D adds a paired ablation of adaptive against static; Section V-E reports the closed-loop learning results, including the workload at which the loop stayed inert and the two at which the delay-prediction error did not improve. The abstract and the conclusion both state that the gradient-based formulation is untrained.

Section / location: Section III-D (p. 4), Section III-E (p. 4), Section V-D (p. 7), Section V-E (p. 7), Table V (p. 7), abstract (p. 1), conclusion (p. 8), Section VI final paragraph (p. 7).

Status: Addressed by scoping the claim and by adding an evaluation of the mechanism that is actually implemented. The gradient-based rule remains untrained and is labelled as such.

Comment 3: *Independent dispatch is mainly used as the baseline by the comparison, while the claimed advantage over existing joint optimization approaches is based largely on qualitative comparison rather than direct experimental evaluation under the same conditions.*

Response: The baseline is still independent dispatch, and we now say so without hedging. It is described in full: each request served alone, no cross-service batching, no route optimisation, on the same request pools, the same seeds and the same distance model, so the two configurations differ only in whether feasibility-first batching is applied. We did not re-implement AVIS or any other published joint-optimisation algorithm, and the manuscript no longer implies a head-to-head result against them. The literature comparison is now labelled directional in the text itself, and the absence of a re-implemented joint-routing baseline is recorded as a limitation.

Change made: Section IV-C describes the baseline precisely and states that no published algorithm was re-implemented. Section V-F labels the literature comparison as directional and notes that magnitudes are not comparable. A dedicated limitation was added. We additionally assessed whether a simplified joint OR-Tools baseline could be built on the same pools, seeds, fleet and metrics for this revision. It could not: OR-Tools is a hard import of the dispatch pipeline, the project environment supplies only Windows binaries for it, and no substitute could be installed, so neither the baseline nor its validation against the existing request pools was possible. We did not build an unvalidated approximation in its place.

Section / location: Section IV-C (p. 5), Section V-F (p. 7), Section VI paragraph 4 (p. 7).

Status: Partially addressed. The baseline is described accurately and the over-claim is removed. A direct comparison against another joint-optimisation algorithm is NOT EXPERIMENTALLY ADDRESSED.

Comment 4: *Per-service-type or per-request performance results are not provided by the study, so it is unclear whether the reported gains are consistent across passenger, food, and parcel services or are mainly driven by one category.*

Response: We attempted this for the camera-ready and report the outcome exactly. Service type is recorded at the request level: `SimulationRequest.request_type` carries ride, food or parcel, and

Trip.request_ids_json links each trip back to its requests, so the join exists in the schema. What does not exist is any type-aware outcome aggregation: distance, delay, fuel, utilisation and CO2 are computed per trip, and the only per-type quantity written to the result files is the input request mix. Two obstacles then prevented us from producing the breakdown. First, the per-trip database from the 36 runs is not recoverable — the harness recreates its database on every run and that database no longer exists, and no result file retains trip-level records. Re-running the experiments was not possible in the environment available for this revision, because OR-Tools is a hard dependency of the routing stage and could not be executed there. Second, and more fundamental than tooling: at the two larger workloads 83.3% of trips are shared, and a shared trip routinely carries requests of different types. Splitting one shared trip's distance or fuel between a food order and a passenger ride requires a cost-allocation convention that the system does not define, so even a re-run would not yield a per-type efficiency figure without introducing an assumption we would have to invent. We therefore report the request mix, state plainly that outcome metrics are not split by it, and record the gap as a limitation rather than apportioning the aggregates.

Change made: The request mix is reported in Table II and Section IV-A. A limitation states that per-service breakdowns of distance, fuel and delay were not instrumented and that it therefore cannot be said whether gains are even across categories or concentrated in one.

Section / location: Table II (p. 5), Section IV-A (p. 5), Section VI paragraph 2 (p. 7).

Status: NOT EXPERIMENTALLY ADDRESSED. Acknowledged explicitly as a limitation; no per-service result is claimed or implied anywhere in the paper.

Comment 5: *The practical feasibility of combining different service types is not fully validated because driver and customer acceptance of batched multi-purpose trips, along with the real impact of priority settings, is not evaluated.*

Response: No acceptance study was carried out and none is claimed. The manuscript states that driver willingness to take mixed passenger-and-parcel assignments, customer tolerance of a shared vehicle and a bounded detour, and the way priority levels are set in practice are all outside the technical scope of this paper and will materially affect deployed behaviour. We note that priority enters the score as a three-level system-assigned value, so how that value is set is a policy question the evaluation does not touch.

Change made: A limitation covering driver acceptance, customer acceptance and the practical setting of priority levels was added and is stated in the conclusion's framing of what remains.

Section / location: Section VI paragraph 5 (p. 7); Section III-B priority definition, equation (7) (p. 3).

Status: NOT EXPERIMENTALLY ADDRESSED. Acknowledged explicitly as a limitation.

Comment 6: *The manuscript requires thorough English language editing to improve clarity and readability.*

Response: The manuscript has been rewritten rather than edited. Every section was redrafted for the camera-ready version, so the defects in the reviewed text — broken mid-sentence paragraphs, hyphens inserted inside words, mixed British and American spelling, and non-standard cross-references — do not carry over. British spelling is now used consistently, including in the Index Terms. Cross-references use the IEEE form. Every page of the final PDF was inspected for overlapping text, margin overflow, broken equations and stranded headings.

Change made: Full redraft of all sections; consistent British spelling; IEEE-form cross-references; all equations re-typeset with correct delimiters, quantifiers and minus signs; figures regenerated.

Section / location: Whole manuscript.

Status: Addressed.

Reviewer 2

Comment 1: *Elaborate on how the service-specific time-window tolerances are selected in practice.*

Response: This exposed a real mismatch between the manuscript and the system, and we have corrected the manuscript rather than the other way round. The implementation does not use service-specific tolerance windows. It applies a single global maximum allowed delay of 20 minutes across passenger rides, food orders and parcel drops, and that constant is what every reported result was produced under. The per-type window that the earlier version discussed is a generalisation of the formulation, not a configured parameter, and it is now presented as such. We have not supplied numerical per-type values, because none exist in the system.

Change made: Equation (5) is written with the global window W . The paragraph following it states that W is a single global constant of 20 minutes applied to all three service types, that per-type windows are part of the generalised formulation with the Lipschitz property that follows from them, that per-type values are not separately configured in the system evaluated here, and that no per-type result is claimed. The value appears in Table II as a configuration parameter.

Section / location: Section III-B, equation (5) and the two paragraphs that follow it (p. 3); Table II (p. 5).

Status: Addressed by correcting the claim. The paper now describes the parameter the system actually uses.

Comment 2: *Clarify the relationship between the 43-request pilot results and the additional 2,320-trip deployment results.*

Response: We were unable to reconstruct either dataset or the relationship between them from the project record, and we were not willing to offer an explanation we could not support. Both have therefore been removed from the manuscript in full, together with every figure derived from them, including the fuel and emissions totals and the batch-success rate. The camera-ready evaluation rests entirely on the reproducible 36-run experiment described in Section IV, which can be regenerated from the seeds given in the paper.

Change made: All 43-request and 2,320-trip figures were removed. The results section was rebuilt on the seeded 50/100/250/500 workload framework.

Section / location: Sections IV and V replaced in full (pp. 5–7); explicit withdrawal statement in Section IV-C (p. 5).

Status: Addressed by withdrawing the unsupported data rather than by explaining it.

Comment 3: *Include a clearer discussion of the limitations associated with the single-area and small-scale pilot evaluation.*

Response: Section VI is now a concrete limitations section rather than a list of caveats. It covers the synthetic workload and bounded geographic scope, the absence of live traffic data and multi-city validation, the absence of per-service instrumentation, the fleet-bounded throughput at the two larger workloads and the absence of a re-implemented joint-routing baseline, the unevaluated driver and customer acceptance, the data-handling gaps, the external mapping dependency, and the untrained learning extension. Each states what was not done rather than softening it.

Change made: Section VI rewritten as seven specific limitations. The conclusion names the binding ones directly.

Section / location: Section VI (p. 7); conclusion (p. 8).

Status: Addressed.

Comment 4: *Authors should clarify the privacy and data-handling procedure for location information and aggregated operational data.*

Response: The previous manuscript stated that location data is retained only long enough to complete and audit a trip and then aggregated into anonymised statistics. That did not match the implementation and it has been withdrawn. The system stores raw pickup and drop-off coordinates together with textual addresses for the lifetime of the record. There is no automatic anonymisation, no retention or deletion schedule, no field-level encryption and no formal compliance framework. We now state that plainly, describe where the data is used, and identify what would have to be added before any deployment involving real users. No compliance claim of any kind is made.

Change made: The anonymisation and retention sentence was deleted. A data-handling limitation was added stating what is stored, what is not implemented, where the data is used, and what is required before deployment.

Section / location: Section VI paragraph 6 (p. 7).

Status: Addressed by correcting the claim to match the implementation. Stated as a limitation, since the controls themselves are not implemented.

Comment 5: *It is suggested to explain the practical implications of relying on Google Maps API data for route and travel-time estimation.*

Response: The manuscript now separates three things that were previously blurred. First, the implementation calls the Google Maps Distance Matrix API when a key is configured and falls back to a haversine matrix otherwise; the fallback is to a local computation, not to a second commercial provider, and no alternative provider is implemented. Second, the reported experiments were run deliberately with the key unset so that the haversine path is used and the runs reproduce without network access; all 1,869 trips were computed this way and no external call was issued. Third, the practical implications are stated: per-call cost and quota limits that grow quadratically with pool size given pairwise scoring, added network latency in every dispatch cycle, and degraded route quality under slow or stale responses, with the haversine fallback keeping the pipeline running at the cost of road-network and traffic awareness.

Change made: Section IV-B states the API configuration used for the experiments and why. A limitation covers cost, quota, latency, degradation and the nature of the fallback, and states that no alternative commercial provider exists in the implementation.

Section / location: Section IV-B (p. 5); Table II (p. 5); Section VI paragraph 7 (p. 7).

Status: Addressed.

Corrections not arising from a reviewer comment

Two further corrections were made after re-reading the implementation, and we note them here rather than leaving them to be discovered. First, the architecture figure and Section III-C previously described routing as preceding driver assignment. The implementation does the reverse: the driver and vehicle are selected first, and the route is then optimised for that vehicle, because the routing objective prices fuel through the vehicle's own efficiency. Figure 1 was redrawn from the dispatch code and Section III-C now

presents the two stages in the implemented order, with equations (8) and (9) exchanged accordingly. Second, the driver selector uses five weighted factors, not three; equation (9) previously showed only proximity, vehicle type and workload, and now includes rotation fairness and historical completion rate with their configured coefficients.

Author information

One item requires confirmation from the committee rather than a change by us. The acceptance record for ICCES-260 lists the second author as “S. R. Ramaya”, while the accepted manuscript and all correspondence use “S. R. Ramya”. We have not altered the author block on our own initiative: the author names, their order, the designation, the department, the institution, the location and all four e-mail addresses are carried forward unchanged from the accepted version. We would be grateful if the committee could confirm the correct spelling so that the proceedings metadata and the manuscript agree.